3. Side thrust

This refers to the thrust developed between the axles in the longitudinal direction. This thrust is caused by the force of air acting on sideways. By connecting panhard rods on the chassis frame the side thrust could be minimized.

4. Weight of the body

The total weight of the goods and passengers of the vehicle acting on the axle are called as unsprung mass. They act on the axle through the springs. The total weight of the vehicle develops a shear force and bending moment on the axle. For reducing these forces powerful springs must be used.

4.13 Types of rear axle

The rear axle can be classified in to three different types according to the construction of rear axle shaft and wheel hub connections, They are

1. Semi-floating axle
2. Full-floating axle
3. Three quarter floating axle

4.13.1 Semi- floating axle

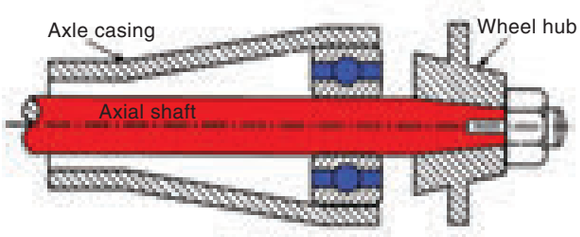


Figure 4.13.1 Semi-floating axle

In this type of axle a bearing is placed in-between the axle shaft and the axle casing. The outer axle shaft is coupled with the wheel hub by using locknut and the key. The axle shaft withstands the vehicle weight, driving thrust and transmits the power to the wheels. It is simple and cheaper compared to other types. This type is used in vehicles such as fiat 100 and standard-20. Figure 4.13.1 shows the view of a semi floating rear axle.

4.13.2 Full floating axle

In this type, one bearing is placed at the differential housing and the two tapper roller bearings are placed between the axle housing and wheel hub. Moreover the axle shaft is kept inside the axle housing and at its other end wheel hub is fitted by using a flange. The axle casing bears the whole weight of the vehicle and axle shaft transmits only the driving torque to the wheels. This type of axle is used in Ashok Leyland tarus, Tata 1612, 1312, Eicher 10.80, 10.90 Mondraj vehicles. Figure 4.13.2 shows the full floating rear axle.

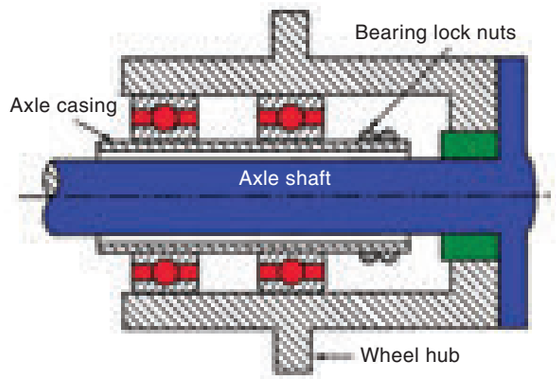


Figure 4.13.2 Full floating axle

Advantages

1. As the axle shaft transmits the power only to wheels the power output is higher.
2. Without removing the wheel and hub the axle can be removed.
3. If the axle shaft is broken, the power transmission only will be stopped.

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